

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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FAMILY TREE

Aim:

To implement a family tree using Prolog and derive relationships between family members.

Objective:

- ☐ To represent parent-child relationships.
- ☐ To derive father, mother, sibling and grandparent relationships.
- ☐ To demonstrate rule-based reasoning.

Algorithm:

- ☐ **Start.**
- ☐ **Define parent facts.**
- ☐ **Define male and female facts.**
- ☐ **Define rules for father, mother, sibling and grandparent.**
- ☐ **Query the required relationship.**
- ☐ **Display the result.**
- ☐ **Stop.**

Flowchart:

START

|

v

Define Family Facts

|

v

Define Relationship Rules

|

v

Enter Query

|

v

Match Facts and Rules

|

v

Display Relationship

|

v

STOP

Prolog Program:

% Family Tree

male(ramesh).

male(arun).

male(kiran).

female(sita).

female(priya).

female(neha).

parent(ramesh, arun).

parent(sita, arun).

parent(ramesh, priya).

parent(sita, priya).

parent(arun, kiran).

parent(priya, neha).

father(X, Y) :-

male(X),

parent(X, Y).

mother(X, Y) :-

female(X),

parent(X, Y).

grandparent(X, Z) :-

parent(X, Y),

parent(Y, Z).

sibling(X, Y) :-

parent(P, X),

parent(P, Y),

X \= Y..

Output:

```
82 ?- father(john, robert).  
true.
```

```
83 ?- mother(mary, robert).  
true.
```

Result:

Thus, the family tree is successfully implemented using Prolog facts and rules.

